

5. Smoking is the largest single cause of avoidable ill health and early death in Wales. In order to reduce this, the Welsh Government has focused on three areas in recent years: discouraging young people from starting to smoke, helping smokers to give up and extending smoke-free environments.

(a) Tobacco smoke contains carcinogens.

State the type of disease caused by carcinogens. [1]

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(b) The table below shows the percentage of the adult population of Wales between 2007 and 2014 who smoked cigarettes.

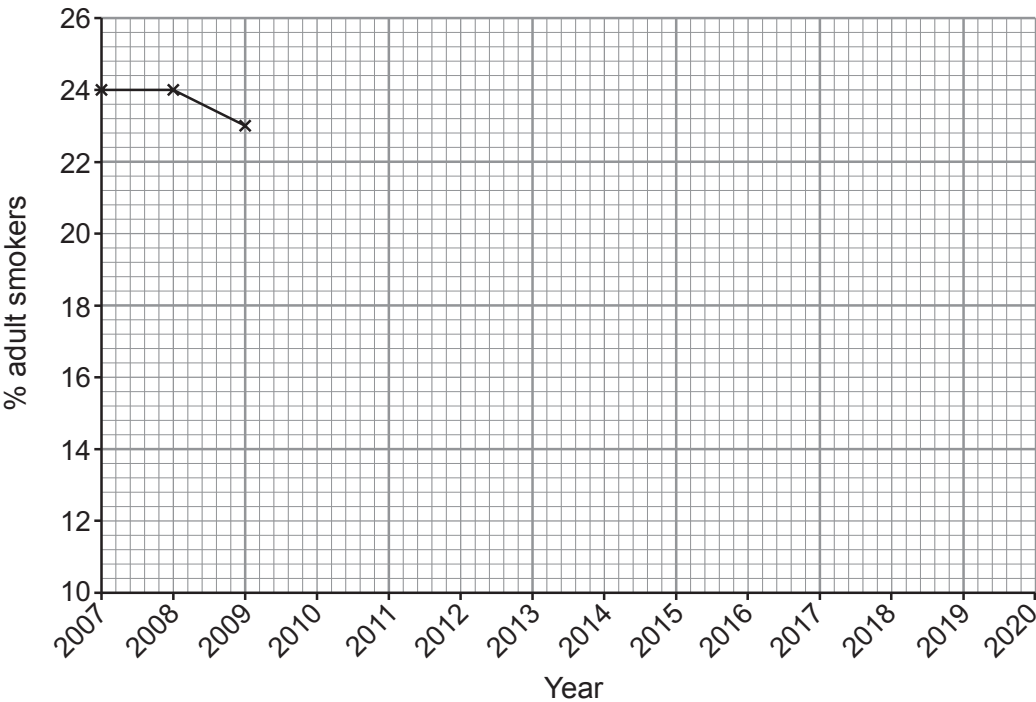
Year	Percentage of the adult population who smoke (%)
2007	24
2008	24
2009	23
2010	22
2011	22
2012	21
2013	20
2014	19

Sources: National Public Health Service for Wales, Welsh Assembly Government, Office for National Statistics

- (i) Complete the graph of the data above on the grid opposite by: [3]
- I. plotting the points from 2010 to 2014
 - II. joining all the plots using a ruler.



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- (ii) In 2007, the Welsh Government banned smoking in enclosed public places as part of a policy to reduce the percentage of adults who smoke.

Give the evidence that the ban had no immediate effect on the percentage of adults who smoke. [1]

- (iii) In 2012, the Welsh Government produced a Tobacco Control Action Plan. One aim was that by 2020 no more than 16% of adults would smoke.

I. **Extend the line on the graph to predict** the % of adult smokers in 2020. Write your answer below. [1]

adult smokers in 2020 = %

II. Your prediction in part (I) is based on an assumption. What does your prediction assume? [1]

- (iv) Suggest **one other** step that could further reduce the percentage of adults who smoke. [1]



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- (c) The breathing rate is the number of times a person breathes in and out in one minute. You have 100 volunteers. Of these, 50 are cigarette smokers and 50 do not smoke.

Plan an investigation to answer the following question:

'Is the breathing rate of smokers different from that of non-smokers?'

Include at least **two** factors which should be kept the same in order to make it a fair test.
[6 QER]

